Graph Sketching: Ellipses

Questions 1–7

Question 1. Sketch each of the following curves, labelled each intersection with the axes clearly.

a) $\frac{x^2}{9} + \frac{y^2}{25} = 1$

b) $\frac{x^2}{16} + \frac{y^2}{4} = 1$

c) $\frac{x^2}{2} + \frac{y^2}{5} = 1$

d) $x^2 + 9y^2 = 36$

e) $x^2 + 5y^2 = 5$

f) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for constants a, b where $a > b$

g) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for constants a, b where $a < b$

h) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for constants a, b where $a = b$

Question 2. Find the vertical tangents of each of the following curves:

a) $3x^2 + 5y^2 = 15$

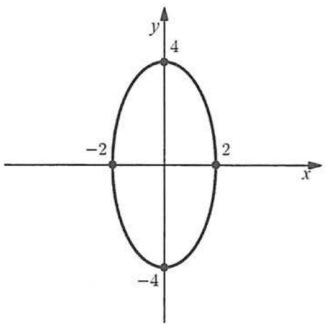
b) $5x^2 + 3y^2 = 15$

Question 3. Each of the following curves are of the form

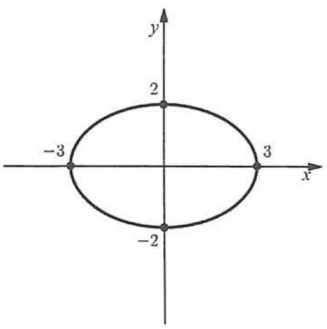
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where a and b are constants. In each case find the value of a and the value of b

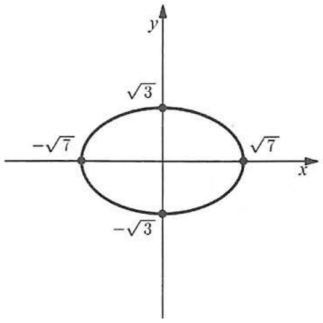
a)



b)



c)



Question 4. For constants a and b, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

It passes through $P(1, 0)$ and $Q(0, -2)$

a) State the value of a and b

b) Sketch the curve including labelling any intersections with the axes.

Question 5. For constants a and b, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

It passes through $P(-\sqrt{6}, 0)$ and $Q(0, 3)$

a) State the value of a and b

b) Sketch the curve including labelling any intersections with the axes.

Question 6. For constants a and b, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

It passes through $P(2, \sqrt{6})$ and $Q(\sqrt{6}, -\sqrt{3})$

a) State the value of a and b

b) Sketch the curve including labelling any intersections with the axes.

Question 7. For a constant a, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{2a} = 1$$

a) Find the points of intersection of the curve with the x-axis.
b) Find the points of intersection of the curve with the y-axis.
c) Sketch the curve including labelling any intersections with the axes in terms of a
d) Given the point $(2, -\sqrt{10})$ lies in the curve, find the value of a